ADICALLY COMPLETE PAIRS ARE HENSELIAN

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The ideal of an adically complete pair is contained in the Jacobson radical. Algebra, Lemma \ref{algebra-lemma-radical-completion}, cited at proof lines 2488-2489. Provides the first half of the henselian property.
- Lifting of factorisations of monic polynomials through surjections with locally nilpotent kernels. Lemma \ref{lemma-locally-nilpotent-henselian}, cited at proof lines 2494-2496. Lifts the factorisation successively modulo the powers of the ideal.

Source reference. Original lemma 0ALJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13372–13372. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 05GI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7062–7062. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0ALI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13371–13371. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let (A, I) be a pair. If A is I-adically complete, then the pair is henselian.*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. By Algebra, Lemma 05GI the ideal I is contained in the Jacobson radical of A. Let $f \in A[T]$ be a monic polynomial and let $\overline{f} = g_0 h_0$ be a factorization of $\overline{f} = f \bmod I$ with $g_0, h_0 \in A/I[T]$ monic generating the unit ideal in $A/I[T]$. By Lemma 0ALI we can successively lift this factorization to $f \bmod I^n = g_n h_n$ with g_n, h_n monic in $A/I^n[T]$ for all $n \geq 1$. As $A = \lim A/I^n$ this finishes the proof. $\square$